

Scrabble Game - HW5

GitHub URL

<https://kathytama.github.io/UML-GUI1/HW5/scrabble.html>

Link to GitHub Repository

[KathyTama/UML-GUI1: GUI1 Assignments](#)

Write-Up: Implemented Features and Status

This project implements a one-line Scrabble game using HTML, CSS, and JavaScript (with jQuery for DOM manipulation and drag-and-drop). The game allows players to drag tiles from a rack onto a board, form words, calculate scores, and validate words against a dictionary. All core features are fully implemented and functional.

Core Features

- **Tile Distribution and Bag:** Tiles are created based on standard Scrabble distribution (A-Z and blank wildcards). The bag is shuffled and tiles are dealt to the player's rack (7 tiles initially).
- **Drag-and-Drop Functionality:** Players can drag tiles from the rack to the board slots. Tiles can be moved back to the rack if needed. Implemented using jQuery UI draggable/droppable.
- **Score Calculation:** Scores are calculated in real-time as tiles are placed, accounting for letter values, double-letter, double-word, and triple bonuses on the board.
- **Word Continuity Check:** Ensures the tiles on the board form a contiguous word (no gaps) before allowing play.
- **Game Flow:** "Play Word" button validates and scores the word, clears the board, deals new tiles, and updates the total score. "Restart" resets the game.
- **UI and Styling:** Clean interface with a background image, styled board, rack, tiles, and controls. Messages are displayed in the message area with color coding (red for errors, green for success).

Extra Credit Features

- **Dictionary Validation:** Words are checked against a loaded dictionary ([dictionary.txt](#)) for validity. If the file is not found, validation is disabled but the game remains playable. Supports wildcards (blanks) by checking all possible letter substitutions.
- **Wildcard Support:** Blank tiles () can represent any letter. Validation uses a recursive function to test all combinations against the dictionary.

Status

- All features are implemented and tested. The game runs in a web browser.

- Dictionary validation requires [dictionary.txt](#) in the project root (download from a word list source like <https://raw.githubusercontent.com/dwyl/english-words/master/words.txt> for full functionality).
- No known bugs; drag-and-drop, scoring, and validation work as expected.
- Dependencies: jQuery (included via CDN in HTML).